

HEMODYNAMIC EQUATIONS AND FORMULAS

1. Cardiac Output and Shunt Determination (see Chapter 3).

A. Fick's Principle.

"The total uptake or release of a substance by an organ is the product of the blood flow to the organ and the arteriovenous concentration difference of that substance."

B. Fick's Method for Determining Cardiac Output.

$$\text{Pulmonary blood flow} = \frac{\text{Oxygen consumption}}{\text{Arteriovenous oxygen difference}}$$

The Fick method is based on Fick's principle. In the absence of a shunt, pulmonary blood flow equals systemic blood flow. The arteriovenous oxygen difference is equal to arterial saturation minus mixed venous saturation. Oxygen content is determined as the product of blood oxygen saturation, the hemoglobin concentration (in grams per deciliter), and the amount of oxygen carried per gram of hemoglobin (Hgb) (1.36 mL oxygen per g of Hgb). This value is multiplied by 10 to correct the units. Thus Fick's formula for determining cardiac output is as follows:

$$\text{Cardiac output} = \frac{\text{Oxygen consumption}}{(\text{Arterial saturation} - \text{mixed venus saturation}) \times (\text{Hgb}) \times 13.6}$$

Note: Oxygen consumption is often assumed (125 mL/min/m² for average individuals and 110 mL/min/m² for elderly patients).

C. Shunt Calculation.

Left-to-right shunt size is described as the ratio of pulmonary to systemic flow, or Qp/Qs. The absolute flow for each circuit can be calculated using the Fick method, or the relationship can be simplified as follows:

$$\text{Qp/Qs} = \frac{\text{Systemic arterial saturation} - \text{mixed venous saturation}}{\text{Pulmonary venous saturation} - \text{pulmonary artery saturation}}$$

In the presence of an atrial septal defect, the empiric "Flamm formula" is used to define the mixed venous saturation as follows:

$$\text{Mixed venous oxygen content} = \frac{3(\text{SVC} - \text{CO}_2 \text{ content}) + 1(\text{IVC} - \text{CO}_2 \text{ content})}{4},$$

where SVC is the superior vena cava and IVC is the inferior vena cava.

A Qp/Qs ratio of 1.3 to 1 is the minimally detectable shunt.

A Qp/Qs ratio of 1.0 to 1.5 indicates a small left-to-right shunt.

A Qp/Qs greater than 2.0 indicates a large left-to-right shunt.

A Qp/Qs less than 1.0 indicates a net right-to-left shunt.

2. Calculation of Valve Orifice Area (see Chapters 5 and 6).

A. The Gorlin Formula for Mitral Valve Area.

$$\text{Mitral valve area} = \frac{\text{Cardiac output}}{(\text{Diastolic filling period}) (\text{heart rate}) (37.9) (\sqrt{\text{Pressure gradient}})}$$

The mitral valve area is in centimeters squared, the cardiac output is in liters per minute, the heart rate is in beats per minute, and the pressure gradient is the mean gradient across the mitral valve in millimeters of mercury.

B. The Gorlin Formula for Aortic Valve Area.

$$\text{Aortic valve area} = \frac{\text{Cardiac output}}{\text{Heart rate (SEP)} (44.5) C (\sqrt{P_1 - P_2})},$$

where SEP is the systolic ejection period.

The aortic valve area is in centimeters squared, the cardiac output is in liters per minute, the heart rate is in beats per minute, and the pressure gradient is the mean gradient across the aortic valve in millimeters of mercury.

C. The Hakki Formula for Aortic Valve Area.

$$\text{AVA} = \frac{\text{Cardiac output in L/min}}{\sqrt{\text{Pressure gradient}}},$$

where AVA is the aortic valve area in centimeters squared, the cardiac output is in liters per minute, and the pressure gradient is the peak-to-peak systolic pressure gradient across the aortic valve.

D. Echocardiography-Derived Formula for Aortic Valve Area.

Bernoulli's Equation:

$$\text{Peak pressure gradient} = 4 \times (\text{Velocity across the aortic valve})^2$$

Aortic valve area is calculated using the continuity equation, which states that the volume of blood passing through the left ventricular outflow tract (LVOT stroke volume) is the same as the volume of blood passing through the aortic valve orifice (AV stroke volume). In turn, stroke volume in a tube can be determined by the formula:

$$\text{Stroke volume} = (\text{Cross-sectional area of orifice}) \times (\text{velocity time integral [VTI]})$$

Assuming a round orifice, the cross-sectional area (CSA) of the LVOT can be calculated by the following equation:

$$\text{CSA LVOT} = \pi \times (\text{LVOT diameter}/2)^2$$

Because the continuity equation states that AV stroke volume equals LVOT stroke volume, the aortic valve area (AVA) can be calculated by the equation:

$$\text{AVA} = \frac{(\text{LVOT VTI}) \times (\text{LVOT CSA})}{\text{AV VTI}}$$

3. Determination of the Regurgitant Fraction in Aortic Regurgitation by Angiography (see [Chapter 6](#)).

$$\text{Regurgitant fraction} = \left(\text{Stroke volume}_{\text{angiography}} \right) - \frac{\text{Stroke volume}_{\text{forward}}}{\text{Stroke volume}_{\text{angiography}}}$$

4. Determination of the Aortic Regurgitation Index in Patients Undergoing Transcatheter Aortic Valve Replacement (see [Chapter 6](#)).

$$\text{AR index} = \frac{\text{Diastolic blood pressure} - \text{left ventricular end-diastolic pressure}}{\text{Systolic blood pressure}},$$

where AR is the aortic regurgitation.

The AR index relates to severity of aortic regurgitation; the lower the index, the greater the degree of paravalvular regurgitation. An AR index of less than 25 is associated with poorer outcomes after transcatheter aortic valve replacement at 1 year.

5. Pulmonary Artery Pressure Estimation by Echocardiography (see [Chapter 9](#)).

$$\text{PA systolic pressure} = 4 (\text{TR V max})^2 + \text{RA pressure},$$

where PA is the pulmonary artery, TR V max is the Doppler velocity of the tricuspid regurgitant jet in meters per second, and RA pressure is the estimated right atrial pressure in millimeters of mercury.

6. Calculation of Pulmonary Vascular Resistance (see [Chapter 9](#)).

$$\text{PVR} = \frac{\text{Mean PA pressure} - \text{mean PCWP}}{\text{Cardiac output (L/min)}},$$

where PVR is pulmonary vascular resistance in Woods units, and the mean pulmonary artery (PA) pressure and the mean pulmonary capillary wedge pressure (PCWP) are in millimeters of mercury. Multiply Woods units $\times 80$ to get resistance units of dyn per s/cm⁵.

7. Pulsatility Index (see [Chapters 9 and 11](#)).

$$\text{PAPi} = \frac{\text{PA systolic pressure in mm Hg} - \text{PA diastolic pressure in mm Hg}}{\text{Right atrial pressure in mm Hg}},$$

where PAPi is the pulmonary artery pulsatility index.
The PAPi is a predictor of right ventricular failure.

8. Diastolic Pressure Gradient (see [Chapter 9](#)).

$$\text{DPG} = \text{PA diastolic pressure} - \text{pulmonary capillary wedge pressure},$$

where DPG is the diastolic pressure gradient and PA is the pulmonary artery.
An elevated DPG of greater than or equal to 7 mm Hg is associated with a worse long-term survival rate in patients with mixed precapillary and postcapillary pulmonary hypertension.

9. Transpulmonary Gradient (see [Chapter 9](#)).

$$\text{TPG} = \text{Mean PA pressure} - \text{mean PCWP},$$

where TPG is the transpulmonary gradient, PA is the pulmonary artery, and PCWP is the pulmonary capillary wedge pressure.

10. Right Ventricular (RV) Stroke Work Index (see [Chapter 9](#))

$$\text{RV Stroke Work Index} = \frac{(\text{mPAP} - \text{RAP}) \times \text{CI} \times 0.0136}{\text{Heart Rate}}$$

where mPAP is the mean pulmonary artery pressure, RAP is the mean right atrial pressure, and CI is the Cardiac Index.

11. Pulmonary Artery (PA) Compliance (see [Chapter 9](#))

$$\text{PA Compliance} = \frac{\text{SV}}{\text{PASP} - \text{PADP}}$$

where SV is the stroke volume, PASP is the PA systolic pressure, and PADP is the PA diastolic pressure.

12. Total Pulmonary Resistance (see [Chapter 9](#))

$$\text{Total Pulmonary Resistance} = \frac{\text{Mean pulmonary artery pressure}}{\text{Cardiac output}}$$

13. Systemic Vascular Resistance (see [Chapter 11](#)).

$$SVR = \frac{\text{Mean aortic pressure} - \text{mean right atrial pressure}}{\text{Cardiac output}},$$

where SVR is the systemic vascular resistance.

Multiplying the SVR by 80 converts to resistance units dyn/s/cm⁵. This can be indexed to body size by multiplying by body surface area.

$$SVRI = SVR \times \text{body surface area},$$

where SVRI is the systemic vascular resistance index.

14. Cardiac Power Output (see [Chapter 11](#)).

$$CPO = \frac{\text{Cardiac output} \times \text{mean arterial pressure}}{451},$$

where CPO is the cardiac power output.

15. Fractional Flow Reserve, Simplified Formula (see [Chapter 14](#)).

$$FFR = \frac{\text{Mean hyperemic distal intracoronary pressure}}{\text{Mean hyperemic aortic pressure}},$$

where FFR is the fractional flow reserve.

16. Fractional Flow Reserve (see [Chapter 14](#)).

$$FFR = \frac{(Pd - Pv)/R_2 \text{ under maximum hyperemia}}{(Pa - Pv)/R_1 \text{ under maximum hyperemia}},$$

where FFR is the fractional flow reserve, Pd is the distal coronary pressure, Pv is the coronary venous pressure, Pa is the aortic pressure, and R is the resistance.